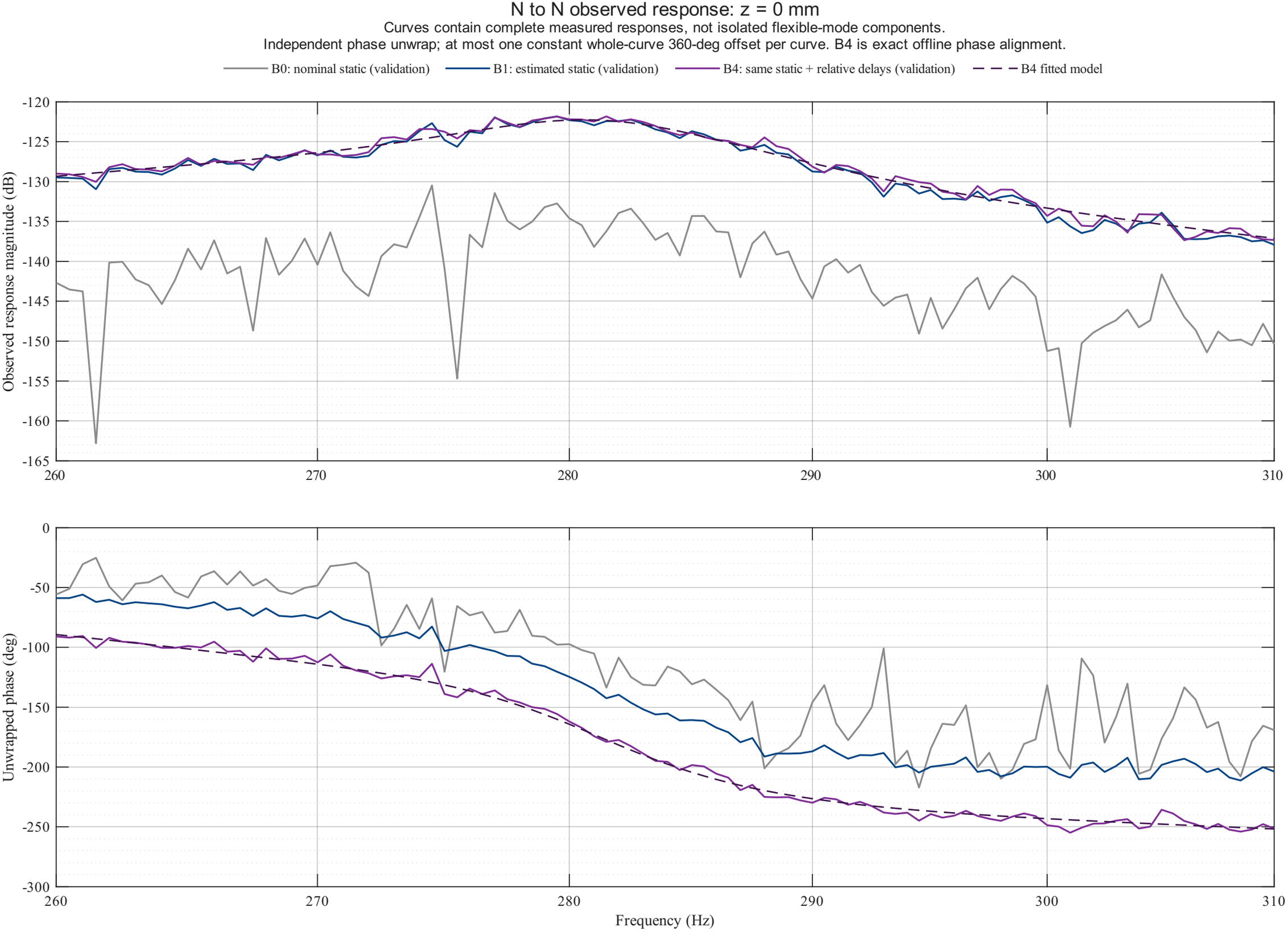

N to N observed response: z = 0 mm
Curves contain complete measured responses, not isolated flexible-mode components.
Independent phase unwrap; at most one constant whole-curve 360-deg offset per curve. B4 is exact offline phase alignment.
B0: nominal static (validation)
B1: estimated static (validation)
B4: same static + relative delays (validation)
B4 fitted model
Observed response magnitude (dB)
-120
-125
-130
-135
-140
-145
-150
-155
-160
-165
Unwrapped phase (deg)
0
-50
-100
-150
-200
-250
-300
260
270
280
290
300
310
Frequency (Hz)